EPM371s: Automatic Control Systems Design

PID Controller Assignment:

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SEC	1

1- Introduction to PID:

- Proportional–Integral–Derivative (PID) controllers are among the most widely used control algorithms in industrial and engineering systems. Their popularity comes from their simple structure, reliability, and ability to provide effective control for a broad range of dynamic systems such as motors, temperature processes, power electronics, and robotics.

- A PID controller continuously calculates an error signal, defined as the difference between a desired setpoint and the measured process variable. Based on this error, the controller outputs a control signal composed of three components:

1. Proportional (P) term – Reacts to the current error.

2. Integral (I) term – Reacts to the accumulation of past errors.

3. Derivative (D) term – Predicts future error behavior.

By tuning the three gains KP, KI, and KD, a PID controller can achieve fast response, minimal steady-state error, and reduced overshoot.

2- PID Controller Equation:

- In the time domain, the control signal $u(t)$is given by:

$$u(t) = K_P\, e(t) + K_I \int_0^t \quad e(\tau)\, d\tau + K_D \frac{de(t)}{dt}$$

Where:

- $e(t) = r(t) - y(t)$: error between desired and actual output.
- Kp: proportional gain.
- Ki: integral gain.
- Kd: derivative gain.

3- Why PID Controllers Are Important:

- Simplicity: Easy to design and implement.
- Robustness: Works well for many systems without detailed modelling.
- Flexibility: Gains can be tuned for speed, stability, and accuracy.
- Wide Application: Used in industrial automation, robotics, power systems, and process control.

4- Tunning Methods :

Common PID tuning methods include:

- Ziegler–Nichols Method
- Cohen–Coon Method
- Trial and Error

- Software-based optimization (MATLAB, LabVIEW, PLCs)

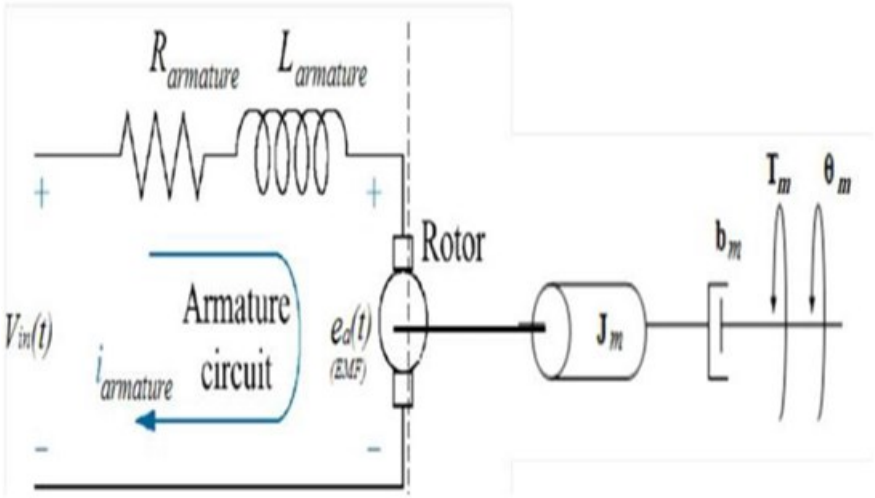
Ziegler–Nichols table:

- A quick and simple way to tune a PID controller.
- Doesn't require a mathematical model of the system.
- Gives good initial parameters that can be refined easily.
- Easy to apply experimentally using $Kcr$and Pcr.
- Widely used in industry and education because it's effective.
- Serves as a baseline to compare other tuning methods.

Ziegler–Nichols Tuning Rule Based on Critical Gain K_{cr} and Critical Period P_{cr} (Second Method)

Type of Controller	K_p	T_i	T_d
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

5- Robotic Arm Example :

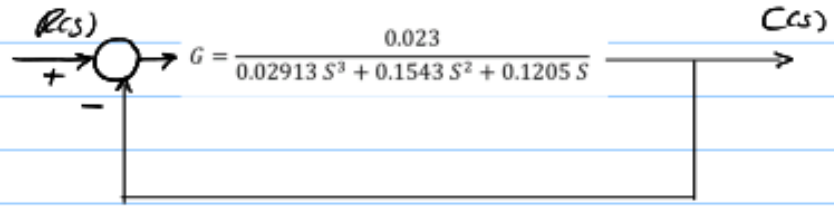


G:

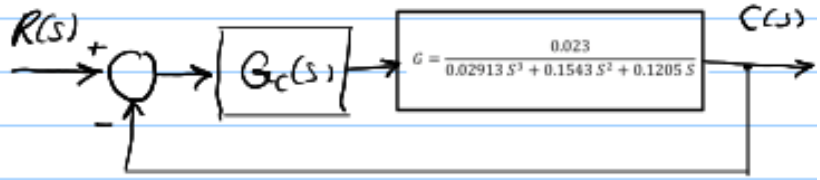
$$G = \frac{K_t * n}{L_a J_{eq} s^3 + (R_a J_{eq} + b_{eq} L_a) s^2 + (R_a b_{eq} + K_t K_b) s}$$

$$G = \frac{0.023}{0.02913\, S^3 + 0.1543\, S^2 + 0.1205\, S}$$

* sys. without any thing



* sys with P-controller



Step 1: By setting $T_i = \infty$ and $T_d = 0$, we obtain the closed-loop transfer function as follows:

$$\frac{C(s)}{R(s)} = \frac{0.023\,k_p}{0.02913\,s^3 + 0.1543\,s^2 + 0.1205\,s + k_p}$$

The characteristic equation for the closed-loop system is

$$0.02913\,s^3 + 0.1543\,s^2 + 0.1205\,s + 0.023\,k_p = 0$$

* Rowth

s^3	0.02913	0.1205
s^2	0.1543	$0.023\,k_p$
s	$\frac{(0.1543)*(0.1205) - (0.02913)*(0.023k_p)}{0.1543}$	0
1	k_p	

$$\therefore \frac{(0.1543)*(0.1205) - (0.02913)*(0.023k_p)}{0.1543} > 0 \rightarrow k_{cr} = 27.75$$

$\therefore \quad 0.1543\, s^2 + 0.023\, k_{cr} = 0$

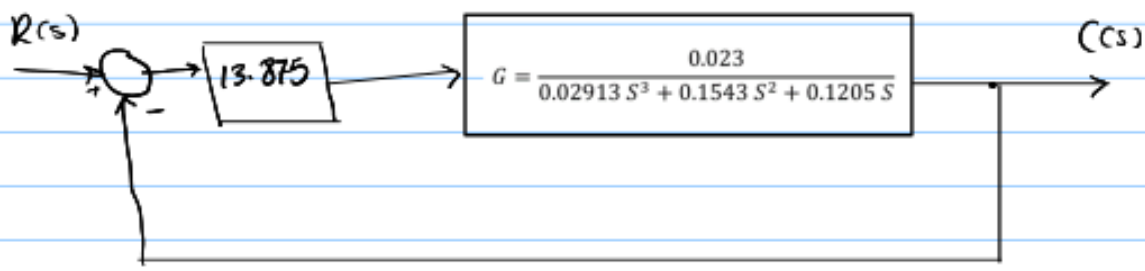
$\therefore \quad 0.1543\, s^2 + 0.023 \times 27.75 = 0$

$\therefore \quad s = \pm j\, 6.4315$

$\therefore \quad P_{cr} = \dfrac{2\pi}{6.4315} = 0.9769$

* for P- controller

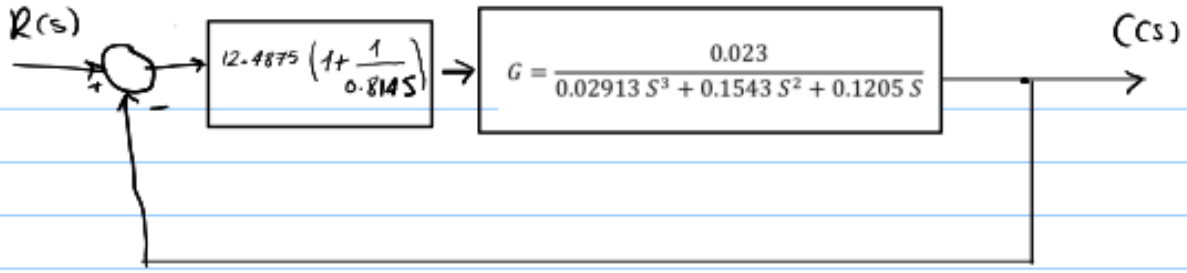
$k_p = 0.5\, k_{cr} = 0.5 \times 27.75 = 13.875$



* for PI Controller :

$k_p = 0.45\, k_{cr} = 0.45 \times 27.75 = 12.4875$

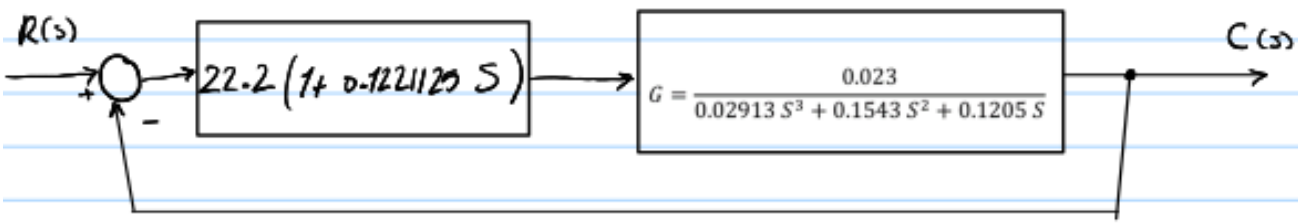
$k_i = \dfrac{P_{cr}}{1.2} = \dfrac{0.9769}{1.2} = 0.814$



* for PD Controller

$$k_p = 0.8\,k_{cr} = 0.8 * 27.75 = 22.2$$

$$k_d = \frac{P_{cr}}{8} = \frac{0.9768}{8} = 0.1221125$$

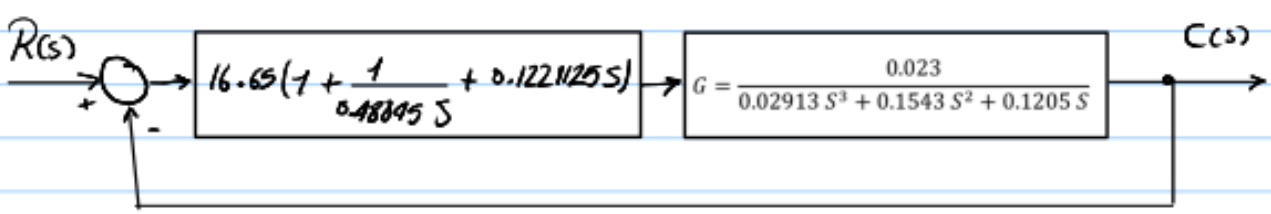


* for PID Controller:

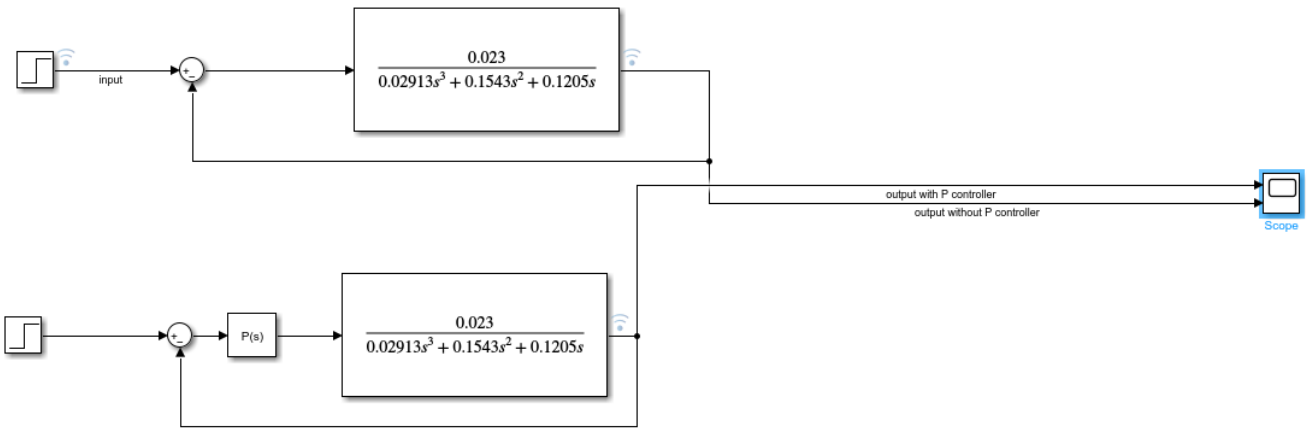
$$k_p = 0.6\,k_{cr} = 0.6 * 27.75 = 16.65$$

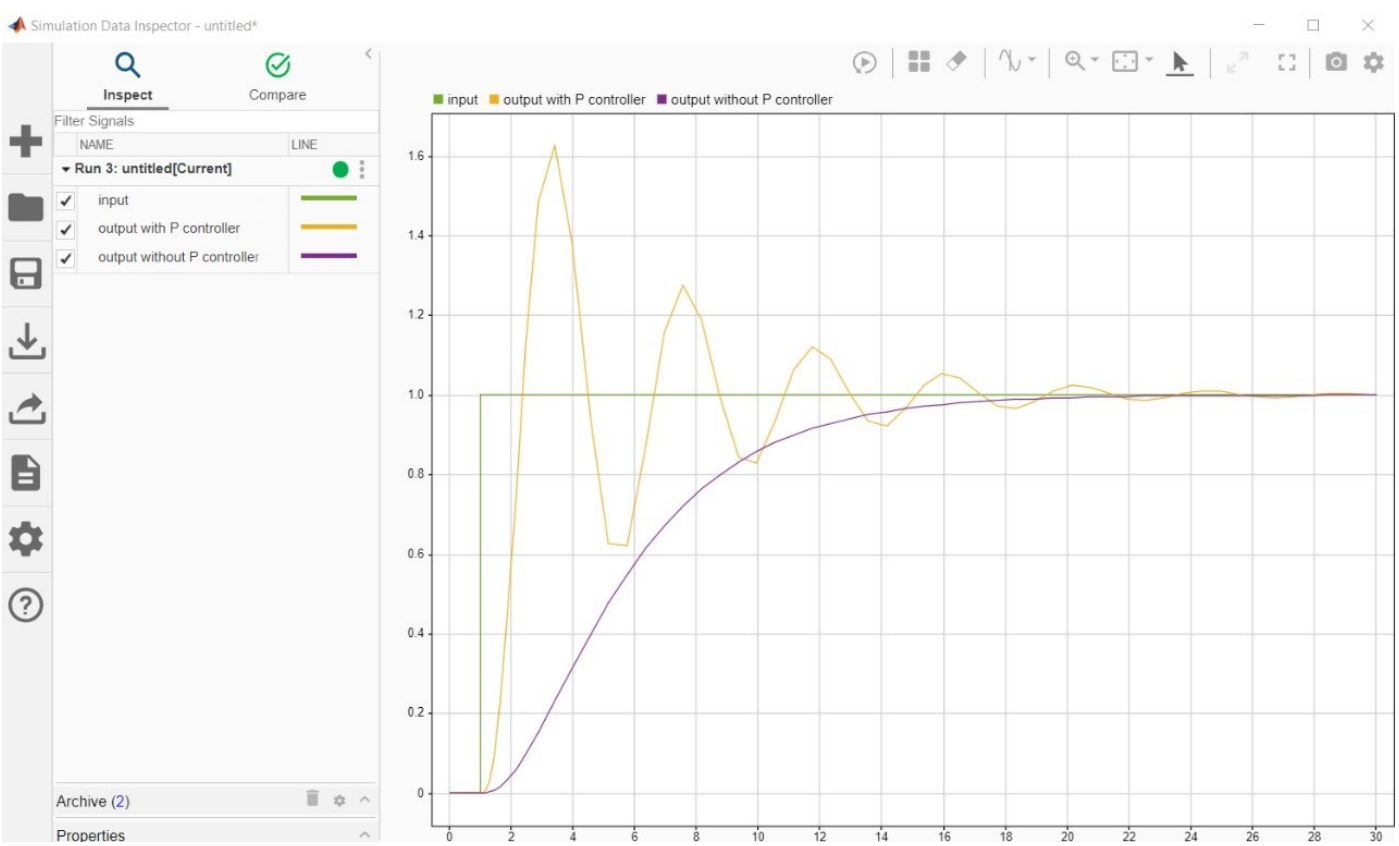
$$k_d = 0.125\,P_{cr} = 0.125 * 0.9769 = 0.1221125$$

$$k_I = 0.5\,P_{cr} = 0.5 \times 0.9769 = 0.48845$$



6-Case (1): P Controller





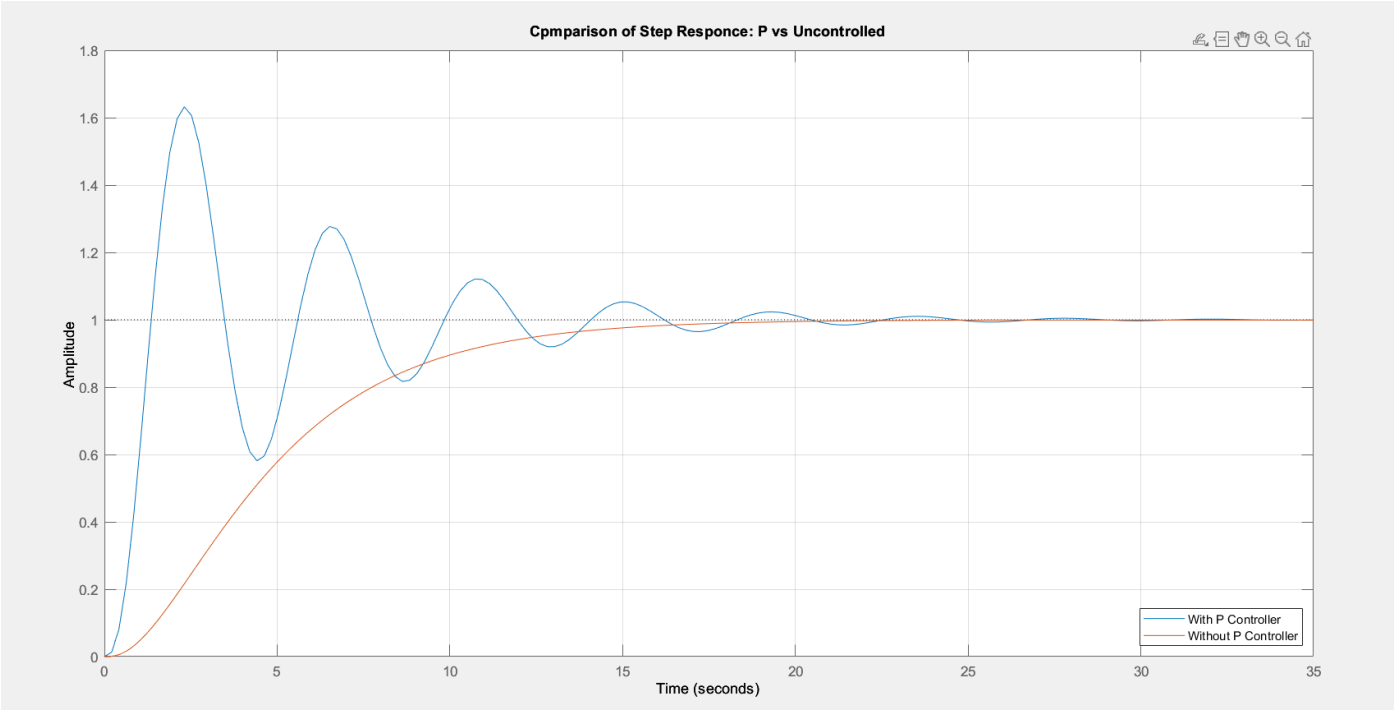
6.1-Code for 1st Case:

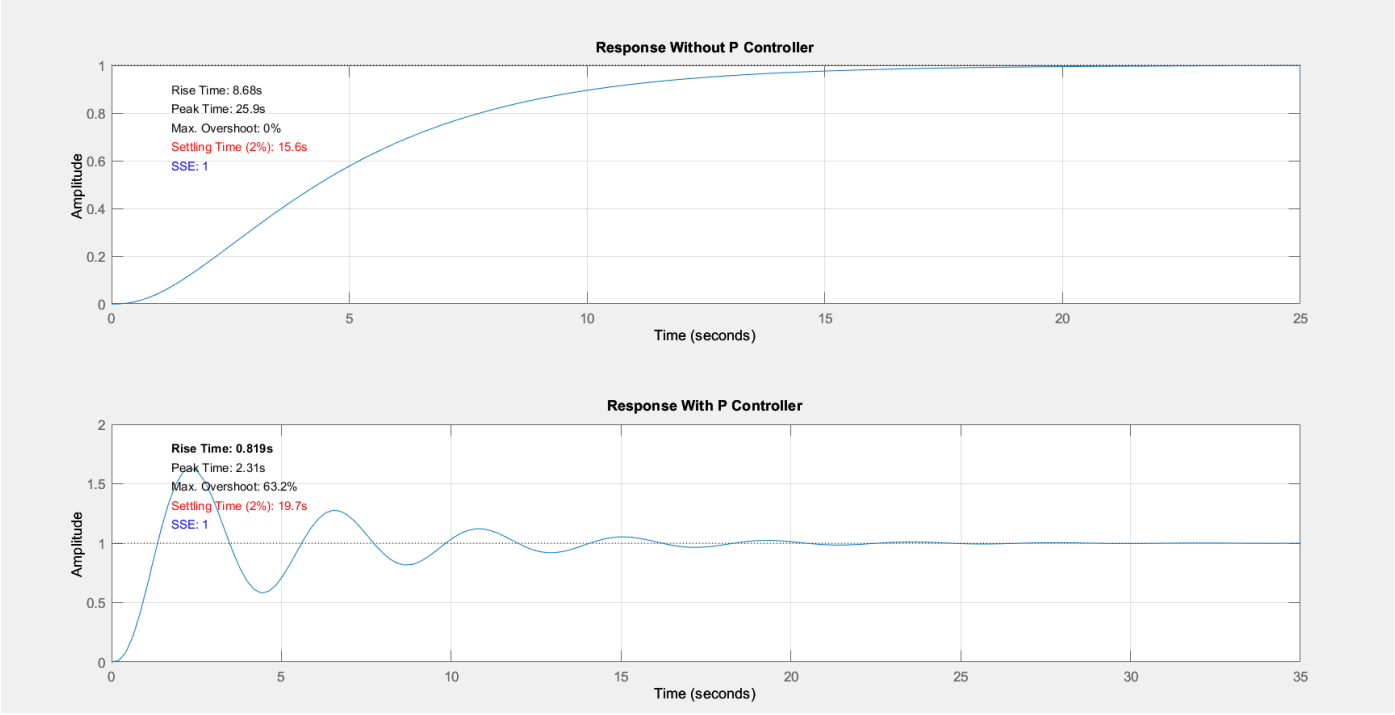
```
s = tf('s');
G = 0.023/(0.02913*s^3 + 0.1543*s^2 + 0.1205*s);
kp = 13.875;

C = pid(kp);
T_pid = feedback(G*C,1);
T_no_pid = feedback(G,1);
S_pid = stepinfo(T_pid);
SteadyStateValue1_pid = dcgain(T_pid);
SteadyStateError1_pid = 1 - SteadyStateValue1_pid;
RiseTime1_pid = ['Rise Time: ',num2str(S_pid.RiseTime,3),'s'];
PeakTime1_pid = ['Peak Time: ',num2str(S_pid.PeakTime,3),'s'];
OverShoot1_pid = ['Max. Overshoot: ',num2str(S_pid.Overshoot,3),'%'];
SettlingTime1_pid = ['Settling Time (2%): ',num2str(S_pid.SettlingTime,3),'s'];
SSE1_pid = ['SSE: ',num2str(SteadyStateValue1_pid,3)];
S2_pid = stepinfo(T_no_pid);
SteadyStateValue2_pid = dcgain(T_no_pid);
SteadyStateError2_pid = 1 - SteadyStateValue2_pid;
RiseTime2_pid = ['Rise Time: ',num2str(S2_pid.RiseTime,3),'s'];
PeakTime2_pid = ['Peak Time: ',num2str(S2_pid.PeakTime,3),'s'];
OverShoot2_pid = ['Max. Overshoot: ',num2str(S2_pid.Overshoot,3),'%'];
SettlingTime2_pid = ['Settling Time (2%): ',num2str(S2_pid.SettlingTime,3),'s'];
SSE2_pid = ['SSE: ',num2str(SteadyStateValue2_pid,3)];
figure;
step(T_pid,T_no_pid);
title('Cpmparison of Step Responce: P vs Uncontrolled');
legend('With P Controller','Without P Controller','Location','SouthEast');
grid on;

figure;
subplot(2,1,1);
step(T_no_pid);
title('Response Without P Controller');
grid on;
ax1 = gca;
XLim1 = get(ax1,'XLim');
YLim1 = get(ax1,'YLim');
X_start1 = XLim1(1) + 0.05 * (XLim1(2) - XLim1(1));
Y_start1 = YLim1(2) - 0.1 * (YLim1(2)-YLim1(1));
line_spacing = 0.08 * (YLim1(2)-YLim1(1));
text(X_start1,Y_start1,RiseTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 1*line_spacing,PeakTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 2*line_spacing,OverShoot2_pid,'FontSize',9);
text(X_start1,Y_start1 - 3*line_spacing,SettlingTime2_pid,'FontSize',9,'Color','red');
text(X_start1,Y_start1 - 4*line_spacing,SSE2_pid,'FontSize',9,'Color','blue');
subplot(2,1,2);
step(T_pid);
title('Response With P Controller');
grid on;
ax2 = gca;
```

```
XLim2 = get(ax2,'XLim');
YLim2 = get(ax2,'YLim');
X_start2 = XLim2(1) + 0.05 * (XLim2(2) - XLim2(1));
Y_start2 = YLim2(2) - 0.1 * (YLim2(2)-YLim2(1));
line_spacing = 0.08 * (YLim2(2)-YLim2(1));
text(X_start2,Y_start2,RiseTime1_pid,'FontSize',9,'FontWeight','bold');
text(X_start2,Y_start2 - 1*line_spacing,PeakTime1_pid,'FontSize',9);
text(X_start2,Y_start2 - 2*line_spacing,OverShoot1_pid,'FontSize',9);
text(X_start2,Y_start2 - 3*line_spacing,SettlingTime1_pid,'FontSize',9,'Color','red');
text(X_start2,Y_start2 - 4*line_spacing,SSE1_pid,'FontSize',9,'Color','blue');
```

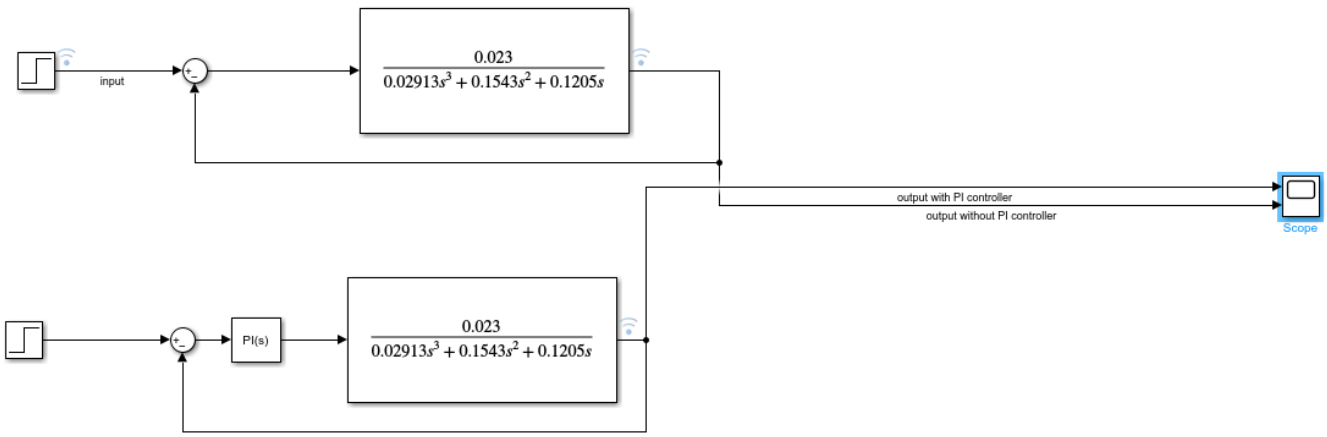


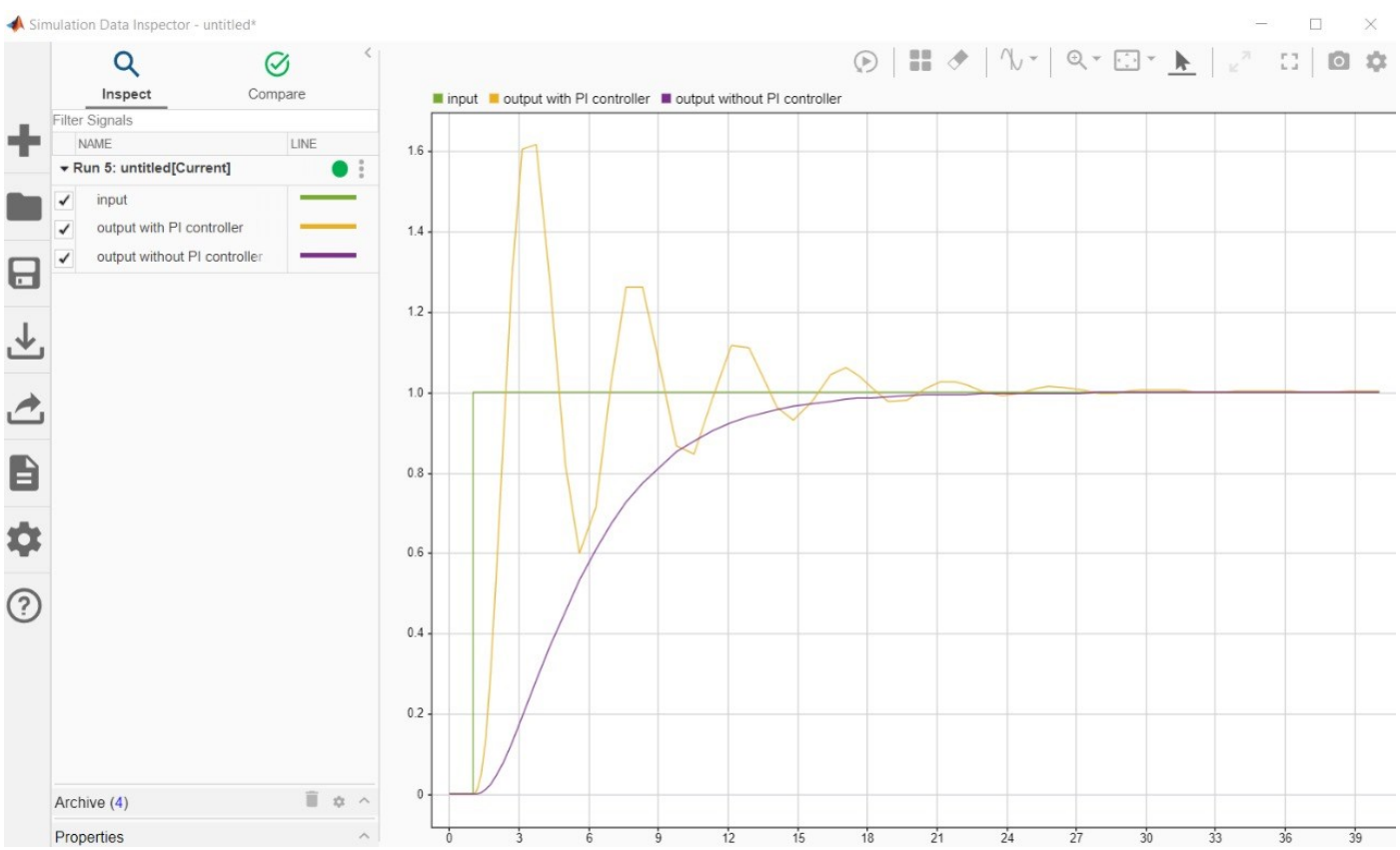


6.2-Comment on 1st Case:

MPOS%	Ts	Tr	Es.s	Tp
Increased	Increased	Decreased	Decreased	Decreased

7-Case (2): PI Controller



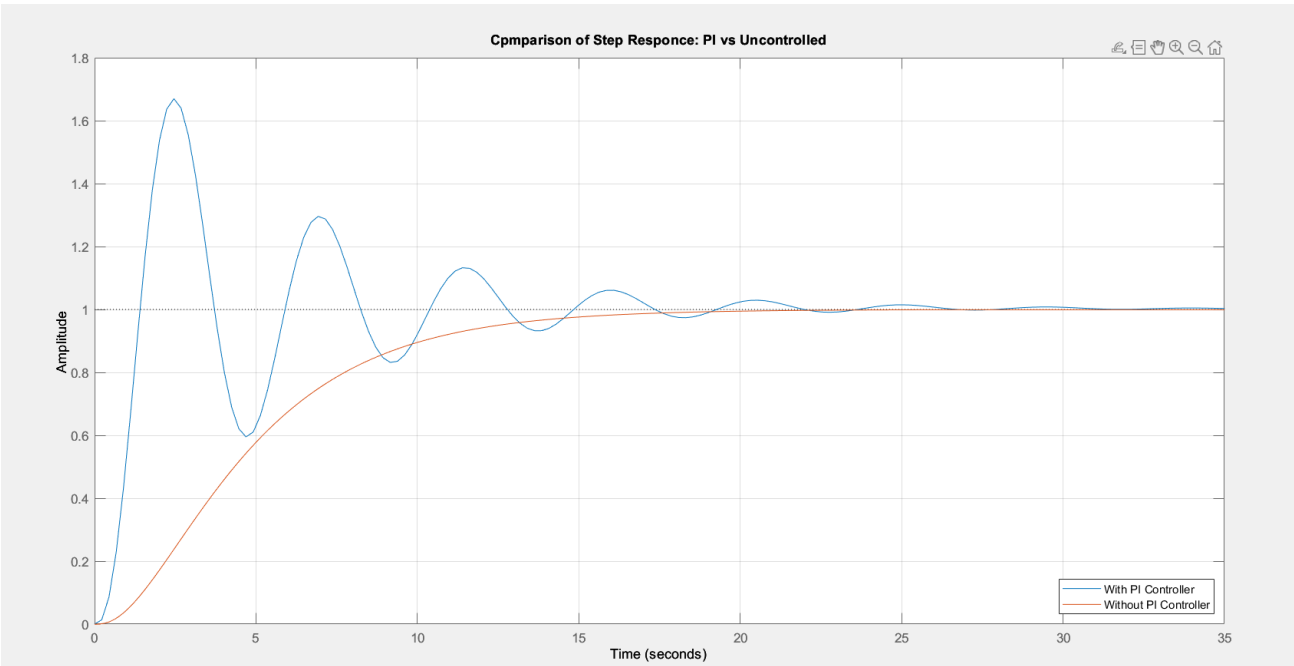


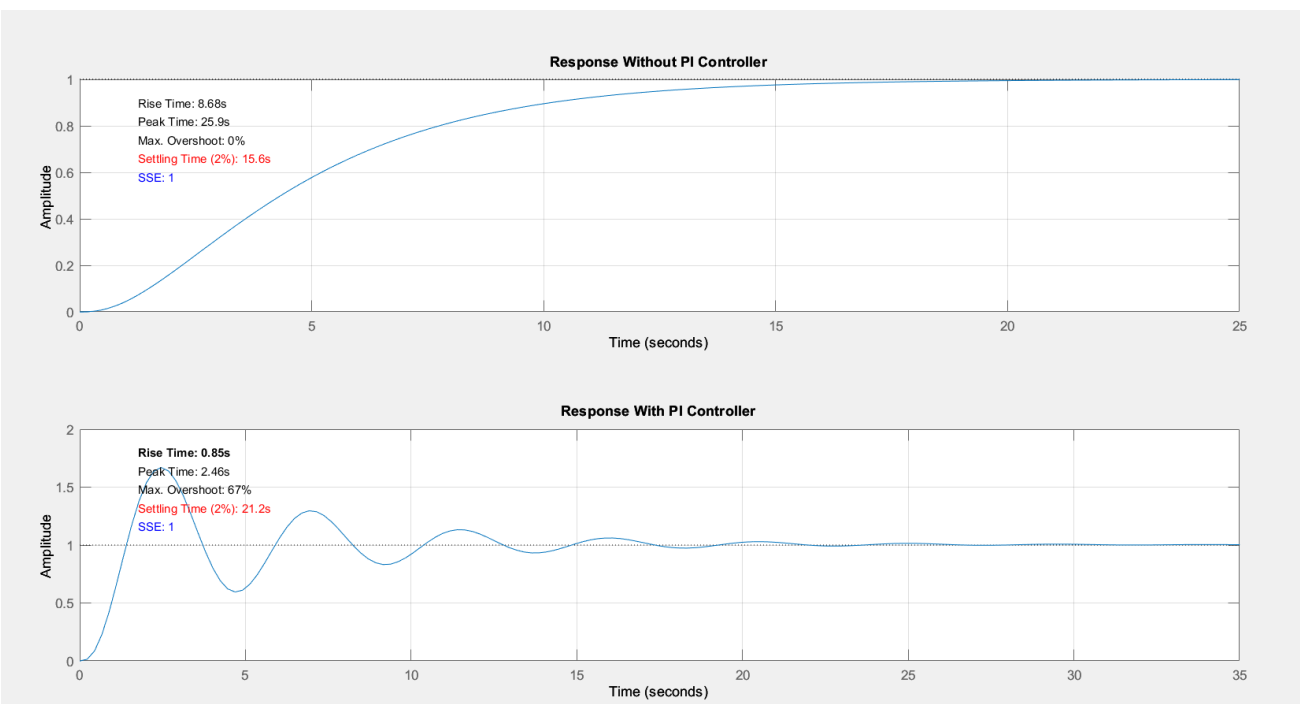
7.1-Code for 2nd Case:

```
s = tf('s');
G = 0.023/(0.02913*s^3 + 0.1543*s^2 + 0.1205*s);
kp = 12.4875;
ki = 0.814;
C = pid(kp,ki);
T_pid = feedback(G*C,1);
T_no_pid = feedback(G,1);
S_pid = stepinfo(T_pid);
SteadyStateValue1_pid = dcgain(T_pid);
SteadyStateError1_pid = 1 - SteadyStateValue1_pid;
RiseTime1_pid = ['Rise Time: ',num2str(S_pid.RiseTime,3),'s'];
PeakTime1_pid = ['Peak Time: ',num2str(S_pid.PeakTime,3),'s'];
OverShoot1_pid = ['Max. Overshoot: ',num2str(S_pid.Overshoot,3),'%'];
SettlingTime1_pid = ['Settling Time (2%): ',num2str(S_pid.SettlingTime,3),'s'];
SSE1_pid = ['SSE: ',num2str(SteadyStateValue1_pid,3)];
S2_pid = stepinfo(T_no_pid);
SteadyStateValue2_pid = dcgain(T_no_pid);
SteadyStateError2_pid = 1 - SteadyStateValue2_pid;
RiseTime2_pid = ['Rise Time: ',num2str(S2_pid.RiseTime,3),'s'];
PeakTime2_pid = ['Peak Time: ',num2str(S2_pid.PeakTime,3),'s'];
OverShoot2_pid = ['Max. Overshoot: ',num2str(S2_pid.Overshoot,3),'%'];
SettlingTime2_pid = ['Settling Time (2%): ',num2str(S2_pid.SettlingTime,3),'s'];
SSE2_pid = ['SSE: ',num2str(SteadyStateValue2_pid,3)];
figure;
step(T_pid,T_no_pid);
title('Cpmparison of Step Responce: PI vs Uncontrolled');
legend('With PI Controller','Without PI Controller','Location','SouthEast');
grid on;

figure;
subplot(2,1,1);
step(T_no_pid);
title('Response Without PI Controller');
grid on;
ax1 = gca;
XLim1 = get(ax1,'XLim');
YLim1 = get(ax1,'YLim');
X_start1 = XLim1(1) + 0.05 * (XLim1(2) - XLim1(1));
Y_start1 = YLim1(2) - 0.1 * (YLim1(2)-YLim1(1));
line_spacing = 0.08 * (YLim1(2)-YLim1(1));
text(X_start1,Y_start1,RiseTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 1*line_spacing,PeakTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 2*line_spacing,OverShoot2_pid,'FontSize',9);
text(X_start1,Y_start1 - 3*line_spacing,SettlingTime2_pid,'FontSize',9,'Color','red');
text(X_start1,Y_start1 - 4*line_spacing,SSE2_pid,'FontSize',9,'Color','blue');
subplot(2,1,2);
step(T_pid);
title('Response With PI Controller');
grid on;
```

```
ax2 = gca;
XLim2 = get(ax2,'XLim');
YLim2 = get(ax2,'YLim');
X_start2 = XLim2(1) + 0.05 * (XLim2(2) - XLim2(1));
Y_start2 = YLim2(2) - 0.1 * (YLim2(2)-YLim2(1));
line_spacing = 0.08 * (YLim2(2)-YLim2(1));
text(X_start2,Y_start2,RiseTime1_pid,'FontSize',9,'FontWeight','bold');
text(X_start2,Y_start2 - 1*line_spacing,PeakTime1_pid,'FontSize',9);
text(X_start2,Y_start2 - 2*line_spacing,OverShoot1_pid,'FontSize',9);
text(X_start2,Y_start2 - 3*line_spacing,SettlingTime1_pid,'FontSize',9,'Color','red');
text(X_start2,Y_start2 - 4*line_spacing,SSE1_pid,'FontSize',9,'Color','blue');
```

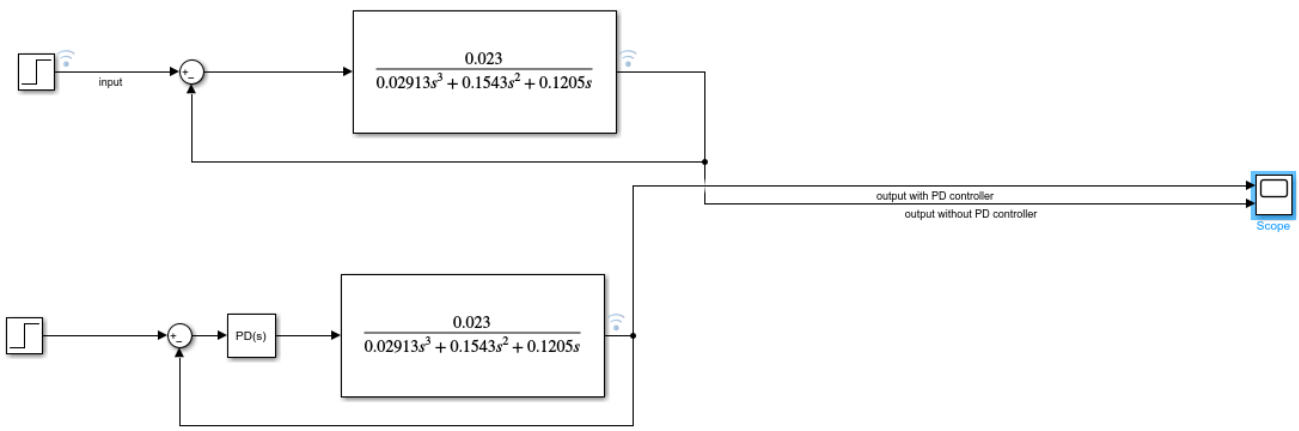


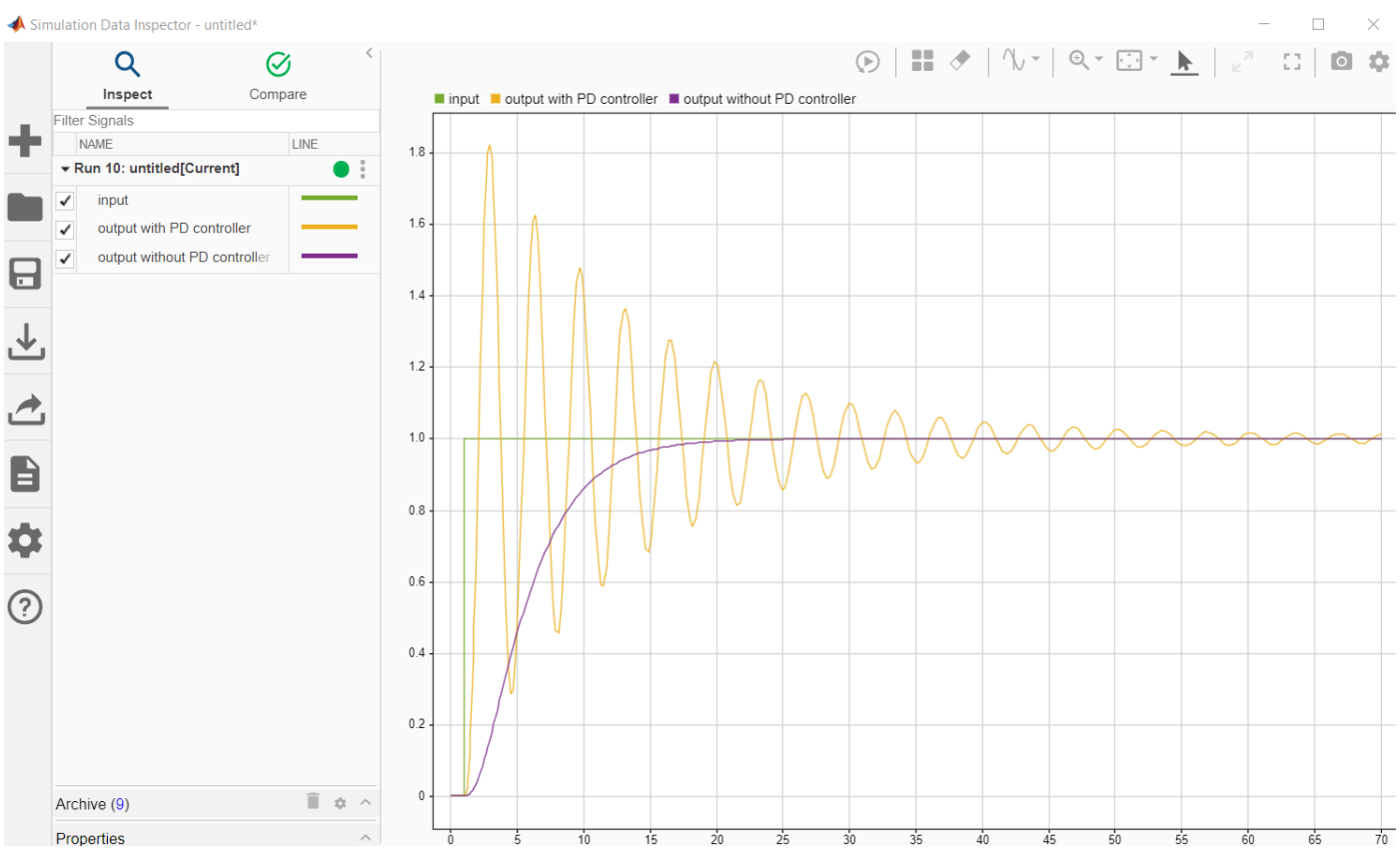


7.2-Comment on 2nd Case:

MPOS%	Ts	Tr	Es.s	Tp
Increased	Increased	Decreased	Decreased	Decreased

8-Case (3): PD Controller



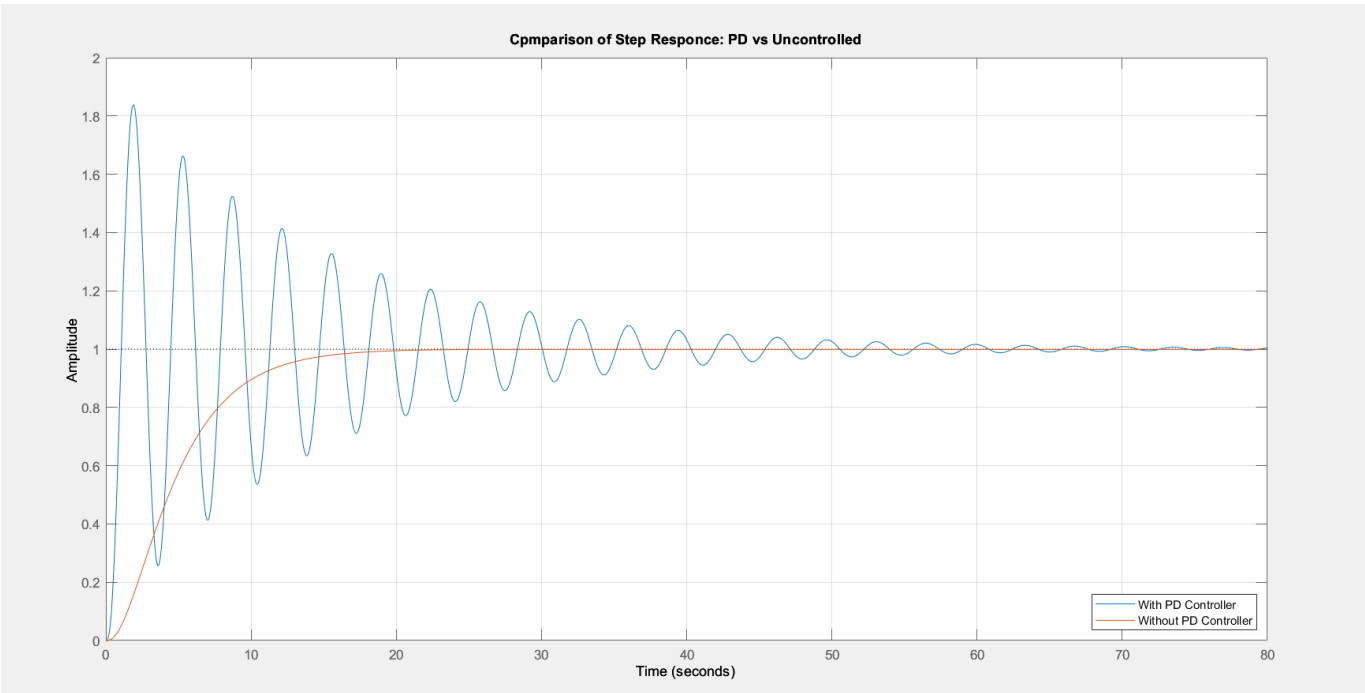


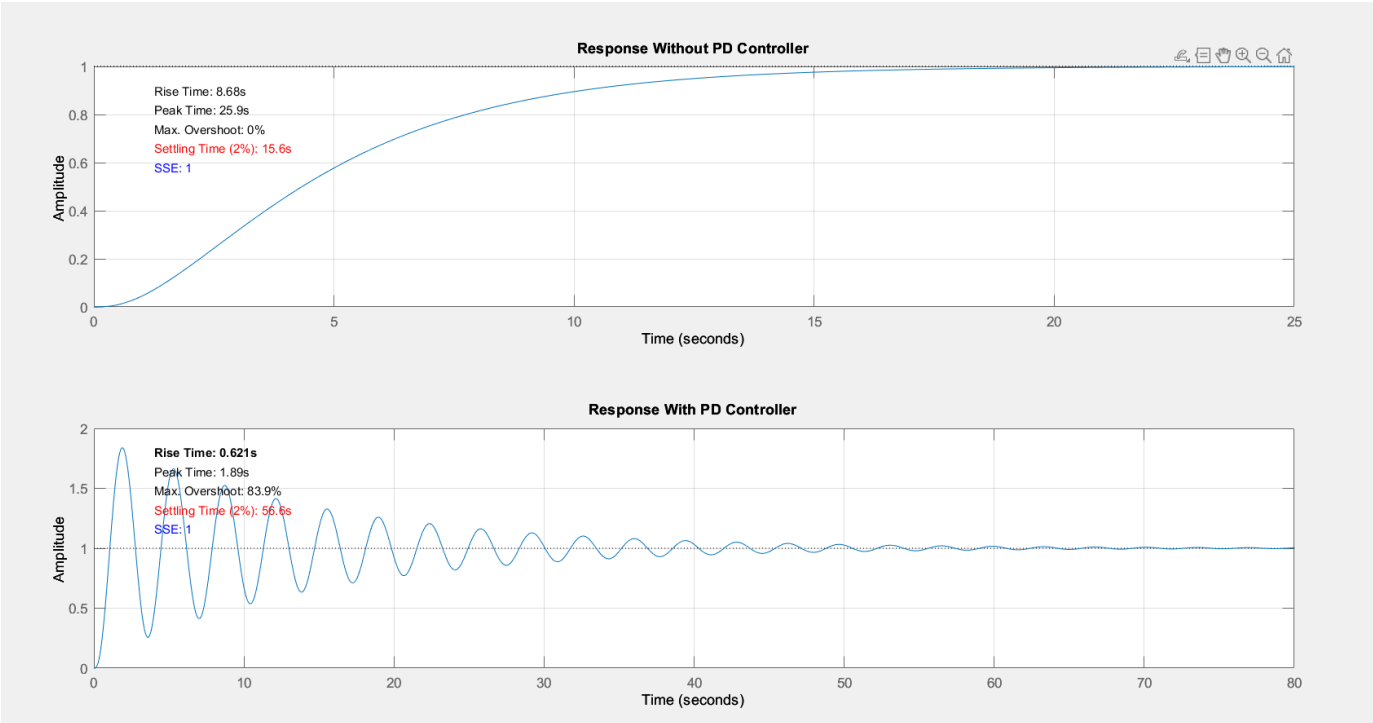
8.1-Code for 4th requirement:

```
s = tf('s');
G = 0.023/(0.02913*s^3 + 0.1543*s^2 + 0.1205*s);
kp = 22.2;
kd = 0.1221125;
C = pid(kp,kd);
T_pid = feedback(G*C,1);
T_no_pid = feedback(G,1);
S_pid = stepinfo(T_pid);
SteadyStateValue1_pid = dcgain(T_pid);
SteadyStateError1_pid = 1 - SteadyStateValue1_pid;
RiseTime1_pid = ['Rise Time: ',num2str(S_pid.RiseTime,3),'s'];
PeakTime1_pid = ['Peak Time: ',num2str(S_pid.PeakTime,3),'s'];
OverShoot1_pid = ['Max. Overshoot: ',num2str(S_pid.Overshoot,3),'%'];
SettlingTime1_pid = ['Settling Time (2%): ',num2str(S_pid.SettlingTime,3),'s'];
SSE1_pid = ['SSE: ',num2str(SteadyStateValue1_pid,3)];
S2_pid = stepinfo(T_no_pid);
SteadyStateValue2_pid = dcgain(T_no_pid);
SteadyStateError2_pid = 1 - SteadyStateValue2_pid;
RiseTime2_pid = ['Rise Time: ',num2str(S2_pid.RiseTime,3),'s'];
PeakTime2_pid = ['Peak Time: ',num2str(S2_pid.PeakTime,3),'s'];
OverShoot2_pid = ['Max. Overshoot: ',num2str(S2_pid.Overshoot,3),'%'];
SettlingTime2_pid = ['Settling Time (2%): ',num2str(S2_pid.SettlingTime,3),'s'];
SSE2_pid = ['SSE: ',num2str(SteadyStateValue2_pid,3)];
figure;
step(T_pid,T_no_pid);
title('Cpmparison of Step Responce: PD vs Uncontrolled');
legend('With PD Controller','Without PD Controller','Location','SouthEast');
grid on;

figure;
subplot(2,1,1);
step(T_no_pid);
title('Response Without PD Controller');
grid on;
ax1 = gca;
XLim1 = get(ax1,'XLim');
YLim1 = get(ax1,'YLim');
X_start1 = XLim1(1) + 0.05 * (XLim1(2) - XLim1(1));
Y_start1 = YLim1(2) - 0.1 * (YLim1(2)-YLim1(1));
line_spacing = 0.08 * (YLim1(2)-YLim1(1));
text(X_start1,Y_start1,RiseTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 1*line_spacing,PeakTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 2*line_spacing,OverShoot2_pid,'FontSize',9);
text(X_start1,Y_start1 - 3*line_spacing,SettlingTime2_pid,'FontSize',9,'Color','red');
text(X_start1,Y_start1 - 4*line_spacing,SSE2_pid,'FontSize',9,'Color','blue');
subplot(2,1,2);
step(T_pid);
title('Response With PD Controller');
grid on;
ax2 = gca;
```

```
XLim2 = get(ax2,'XLim');
YLim2 = get(ax2,'YLim');
X_start2 = XLim2(1) + 0.05 * (XLim2(2) - XLim2(1));
Y_start2 = YLim2(2) - 0.1 * (YLim2(2)-YLim2(1));
line_spacing = 0.08 * (YLim2(2)-YLim2(1));
text(X_start2,Y_start2,RiseTime1_pid,'FontSize',9,'FontWeight','bold');
text(X_start2,Y_start2 - 1*line_spacing,PeakTime1_pid,'FontSize',9);
text(X_start2,Y_start2 - 2*line_spacing,OverShoot1_pid,'FontSize',9);
text(X_start2,Y_start2 - 3*line_spacing,SettlingTime1_pid,'FontSize',9,'Color','red');
text(X_start2,Y_start2 - 4*line_spacing,SSE1_pid,'FontSize',9,'Color','blue');
```

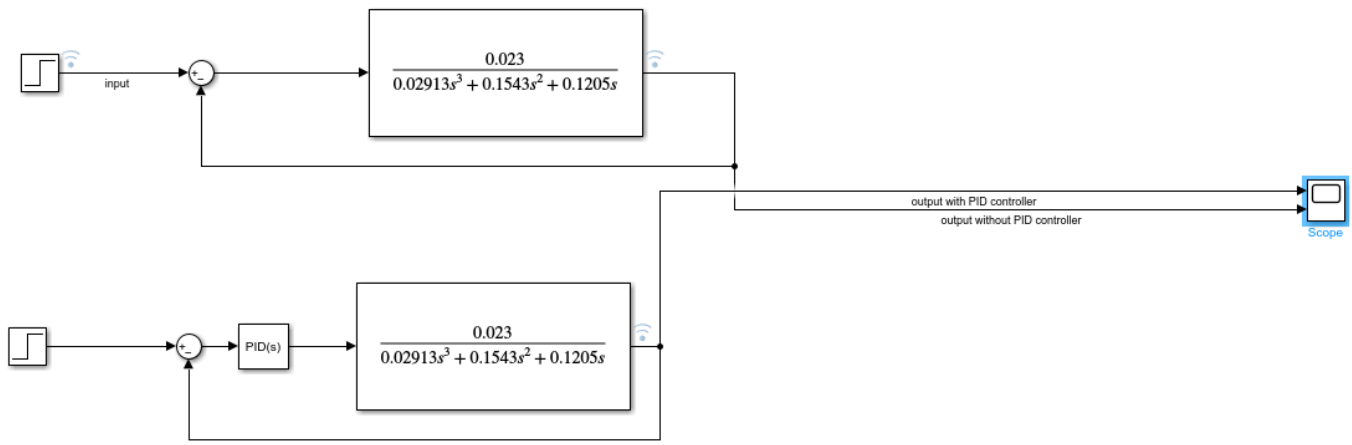


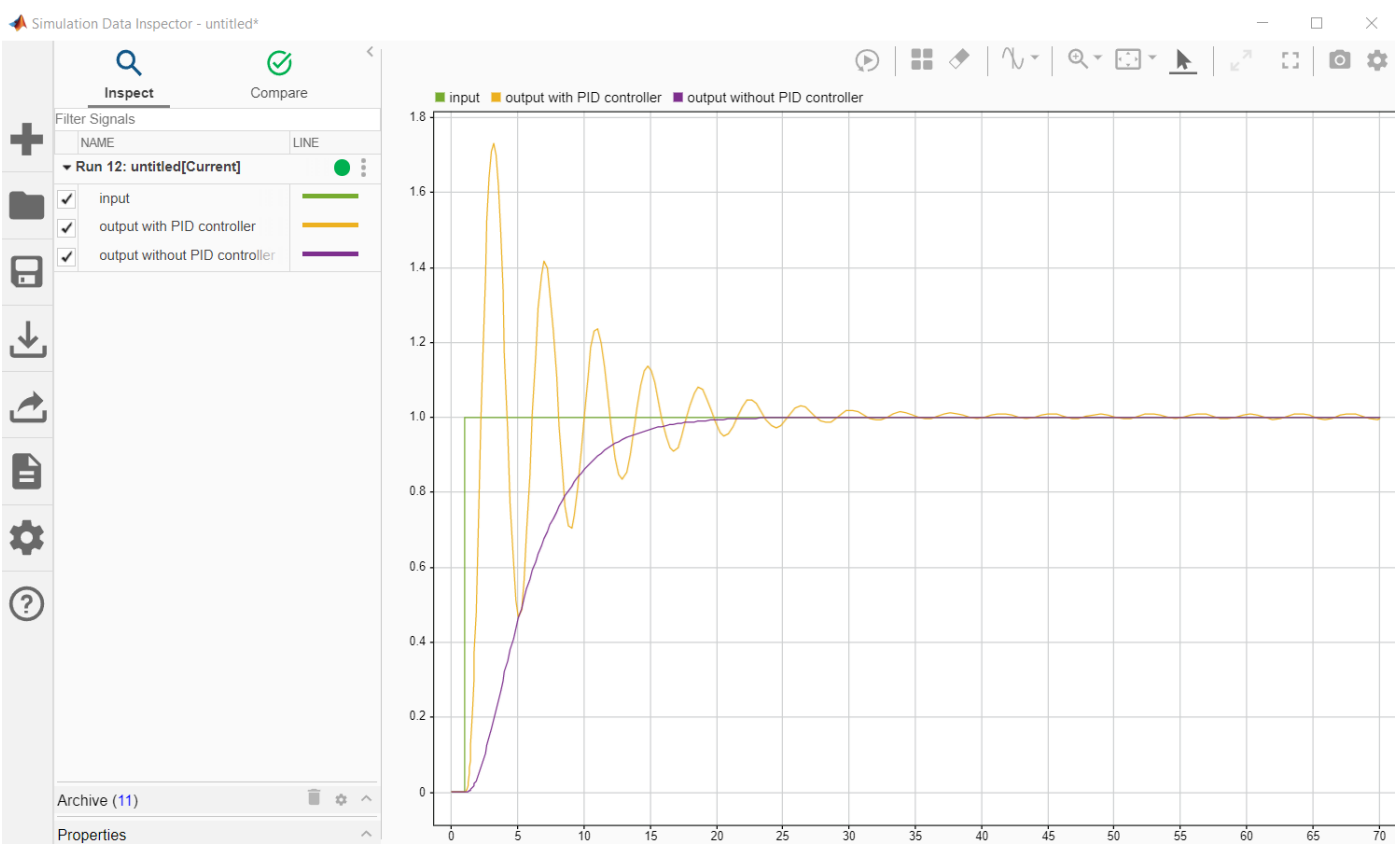


8.2-Comment on 3rd Case:

MPOS%	Ts	Tr	Es.s	Tp
Increased	Increased	Decreased	Decreased	Decreased

9-Case (4): PID Controller



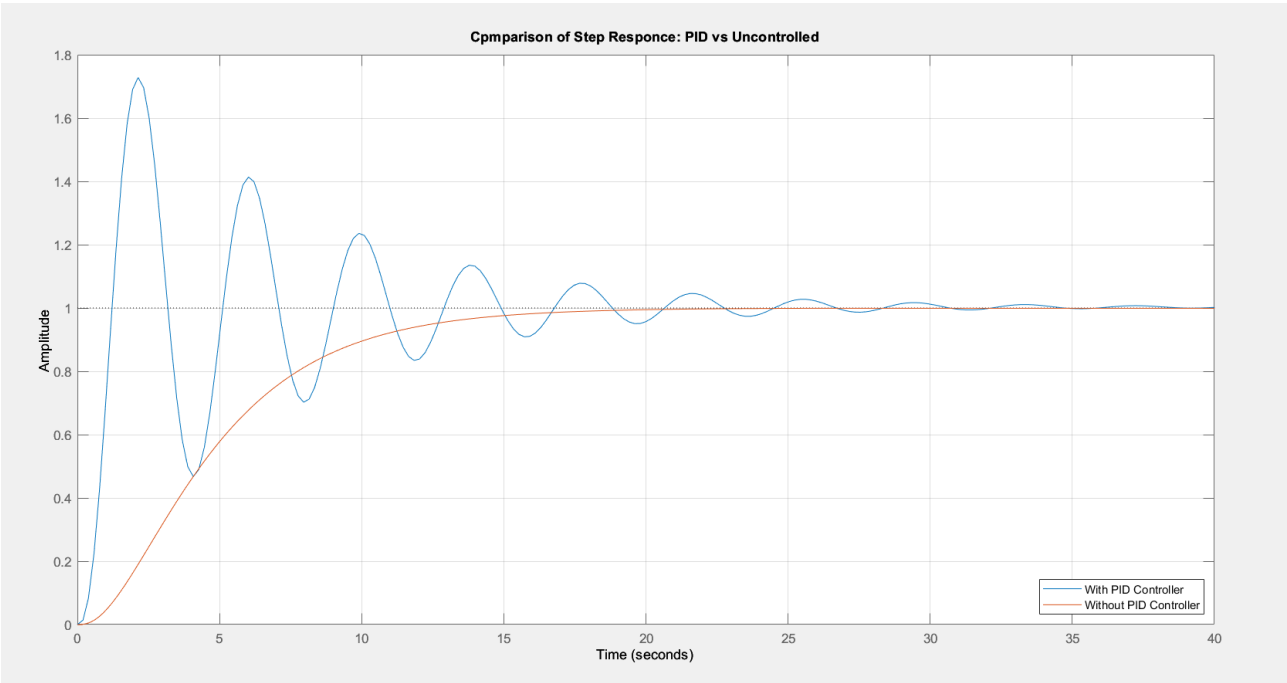


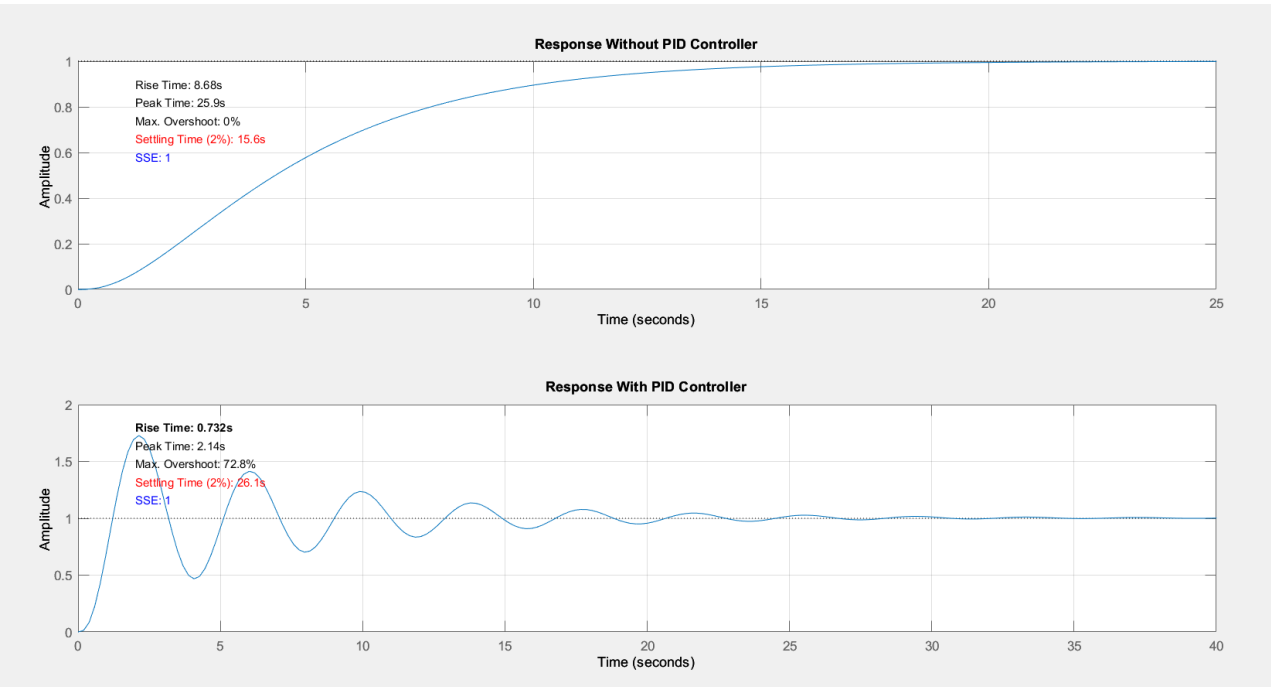
9.1-Code for 5th requirement:

```
s = tf('s');
G = 0.023/(0.02913*s^3 + 0.1543*s^2 + 0.1205*s);
kp = 16.65;
ki = 0.48845;
kd = 0.1221125;
C = pid(kp,ki,kd);
T_pid = feedback(G*C,1);
T_no_pid = feedback(G,1);
S_pid = stepinfo(T_pid);
SteadyStateValue1_pid = dcgain(T_pid);
SteadyStateError1_pid = 1 - SteadyStateValue1_pid;
RiseTime1_pid = ['Rise Time: ',num2str(S_pid.RiseTime,3),'s'];
PeakTime1_pid = ['Peak Time: ',num2str(S_pid.PeakTime,3),'s'];
OverShoot1_pid = ['Max. Overshoot: ',num2str(S_pid.Overshoot,3),'%'];
SettlingTime1_pid = ['Settling Time (2%): ',num2str(S_pid.SettlingTime,3),'s'];
SSE1_pid = ['SSE: ',num2str(SteadyStateValue1_pid,3)];
S2_pid = stepinfo(T_no_pid);
SteadyStateValue2_pid = dcgain(T_no_pid);
SteadyStateError2_pid = 1 - SteadyStateValue2_pid;
RiseTime2_pid = ['Rise Time: ',num2str(S2_pid.RiseTime,3),'s'];
PeakTime2_pid = ['Peak Time: ',num2str(S2_pid.PeakTime,3),'s'];
OverShoot2_pid = ['Max. Overshoot: ',num2str(S2_pid.Overshoot,3),'%'];
SettlingTime2_pid = ['Settling Time (2%): ',num2str(S2_pid.SettlingTime,3),'s'];
SSE2_pid = ['SSE: ',num2str(SteadyStateValue2_pid,3)];
figure;
step(T_pid,T_no_pid);
title('Cpmparison of Step Responce: PID vs Uncontrolled');
legend('With PID Controller','Without PID Controller','Location','SouthEast');
grid on;

figure;
subplot(2,1,1);
step(T_no_pid);
title('Response Without PID Controller');
grid on;
ax1 = gca;
XLim1 = get(ax1,'XLim');
YLim1 = get(ax1,'YLim');
X_start1 = XLim1(1) + 0.05 * (XLim1(2) - XLim1(1));
Y_start1 = YLim1(2) - 0.1 * (YLim1(2)-YLim1(1));
line_spacing = 0.08 * (YLim1(2)-YLim1(1));
text(X_start1,Y_start1,RiseTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 1*line_spacing,PeakTime2_pid,'FontSize',9);
text(X_start1,Y_start1 - 2*line_spacing,OverShoot2_pid,'FontSize',9);
text(X_start1,Y_start1 - 3*line_spacing,SettlingTime2_pid,'FontSize',9,'Color','red');
text(X_start1,Y_start1 - 4*line_spacing,SSE2_pid,'FontSize',9,'Color','blue');
subplot(2,1,2);
step(T_pid);
title('Response With PID Controller');
grid on;
```

```
ax2 = gca;
XLim2 = get(ax2,'XLim');
YLim2 = get(ax2,'YLim');
X_start2 = XLim2(1) + 0.05 * (XLim2(2) - XLim2(1));
Y_start2 = YLim2(2) - 0.1 * (YLim2(2)-YLim2(1));
line_spacing = 0.08 * (YLim2(2)-YLim2(1));
text(X_start2,Y_start2,RiseTime1_pid,'FontSize',9,'FontWeight','bold');
text(X_start2,Y_start2 - 1*line_spacing,PeakTime1_pid,'FontSize',9);
text(X_start2,Y_start2 - 2*line_spacing,OverShoot1_pid,'FontSize',9);
text(X_start2,Y_start2 - 3*line_spacing,SettlingTime1_pid,'FontSize',9,'Color','red');
text(X_start2,Y_start2 - 4*line_spacing,SSE1_pid,'FontSize',9,'Color','blue');
```





9.2-Comment on 4th Case:

MPOS%	Ts	Tr	Es.s	Tp
Increased	Increased	Decreased	Decreased	Decreased